Privacy is Value · V6: The Swordsman Reading — Compressed Specification

Compressed specification of the Privacy Value Model · equations only · a volume of the Privacy is Value book

privacymage

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The Equation

Static form:

$$\boxed{\begin{aligned} V(\pi,t) = P^{1.5} \cdot C \cdot Q \cdot S \cdot e^{-\lambda t} \cdot (1 + A_h(\tau)) \\ \cdot \left(1 + \sum_i w_i \frac{n_i}{N_0}\right)^k \cdot G(\text{guilds}) \\ \cdot R(d, \text{compression}, \rho) \cdot M(u, y) \\ \cdot \Phi_{\text{agent}}(\Sigma) \cdot \Phi_{\text{data}}(\Delta) \cdot \Phi_{\text{inference}}(\Gamma) \\ \cdot T_{\int}(\pi) \end{aligned}}$$

Differential form: $\frac{dV}{dt} = \nabla_{\partial M} \cdot J_{\partial M} + S(x) - D(x)$

Gating: Multiplicative. Any term $= 0 \implies V = 0$.

Lattice: π is a path through $\mathcal{L} = \mathbb{Z}/64\mathbb{Z}$, t is time since data generation.

Grouped V6 reading of the same equation:

$$V(\pi,t) = \Phi_{v5} \cdot \left[\sum_{\text{terms}}\right] \cdot A_h(\tau) \cdot T_{\int}(\pi)$$

$$\Phi_{v5} = \Phi_{agent}(\Sigma) \cdot \Phi_{data}(\Delta) \cdot \Phi_{inference}(\Gamma)$$

Multiplicative gating: any axis at 0 collapses V. The form is unchanged from V5.4. V6 takes the t in the signature seriously everywhere below.

Inherited Terms (V1–V4)

Symbol	Name	Domain	Description
P	Privacy Strength	$[0, 1]$	Cryptographic enforcement. Exponent 1.5 via C6.
C	Credential Verifiability	$[0, 1]$	Verify without revealing.
Q	Data Quality	$[0, 1]$	Accuracy, completeness, fitness.
S	Scope / Sensitivity	$\mathbb{R}^+$	Domain-specific multiplier.
$e^{-\lambda t}$	Temporal Decay	$(0, 1]$	Freshness. $\lambda > 0$.
$M(u, y)$	Market Maturity	$[0, 1]$	User sophistication, market year.

Holonic Temporal Memory

$$A_h(\tau) = \sum_j p(\tau_j) \cdot w(\tau_j) \cdot e^{-\mu \cdot \mathrm{age}(\tau_j)}$$

GUID-addressed holons. Infrastructure-independent. $p(\tau_j) = 0 \implies A_h = 0$.

Three-Axis Separation

$$\Phi_{v5} = \Phi_{\mathrm{agent}}(\Sigma) \cdot \Phi_{\mathrm{data}}(\Delta) \cdot \Phi_{\mathrm{inference}}(\Gamma)$$

Axis	Formula	Meaning
Agent	$\Phi_a = \min(1, \frac{S/M}{\varphi}) \cdot \det(\Sigma)$	Swordsman $\perp$ Mage. $\cong D_{2n}$ (C14, 75%)
Data	$\Phi_d = 1 - \max_j(\text{share}_j)$	No single provider holds majority
Inference	$\Phi_i = 1 - I(\text{model}; \text{executor})$	Generator $\perp$ Solver

Collapse any axis $\implies$ total collapse.

C7 (multiplicative form, 30%) is the falsification frontier. Boundary cases register-bound: partial collapse (alternatives: additive-with-floor, min()); axis correlation under composition (candidate correction: det(Σ) form); time dependence (repair path: the differential form). Structural anchor (C85, 40%): Protection+Delegation → Σ · Memory+Value → Δ · Connection+Computation → Γ.

Reconstruction Difficulty · the Moving Ceiling

Static form (V5.4):

$$R(d, c, \rho) = R_{\text{base}}(d) \cdot \left(1 - \frac{1}{c}\right) \cdot (1 + \alpha \cdot \rho)$$

$\rho = f(\text{traversal depth, duration, intentional transitions})$. Dual: privacy amplifier + agent maturity. **Ceiling (proven):** $R < 1$ under budget constraints. **Error floor (proven):** $P_e \geq 1 - R_{\max}$ via Fano.

The static form is superseded at V6 by the time-indexed form:

$$R(t) = \frac{C_S(t) + C_M(t)}{H(X)}$$

$$t^* = \sup\{\, t : R(t) < 1 \,\}$$

H(X) fixed by the person; capacities evaluated against the strongest adversary class at time t; R(t) non-decreasing under capability growth (C82, 65%). The mechanism is the decoder, not the data. Every static guarantee has a shelf life.

Guild Efficiency

$$G(\text{guilds}) = \prod_g (1 + \text{efficiency}_g \cdot \text{active}_g / \text{total}_g)$$

Shared-parent coordination: O(1) not O(N^2).

Path Integral

$$T_{\int}(\pi) = 1 + \beta \int_\pi F(\gamma)\, d\gamma \;\cong\; 1 + \beta \sum_{i=1}^{n} R(\text{step}_i)$$

$F(\gamma) = \text{resolution\_depth} \cdot \text{fidelity}$. One lap = one cycle. Dragon ($$62 laps) = closure.

Holographic Bound

$$\partial M : \text{96 edges encoding 64 vertices, toroidal topology} \qquad \frac{96}{64} = 1.5 = P^{1.5}$$

C4 RESOLVED. Boundary encodes bulk. dV/dt computes on ∂M, not the 64-vertex interior.

$J_{\partial M} = J_{\text{agent}} + J_{\text{data}} + J_{\text{inference}} + J_{\text{compression}} + J_{\text{holonic}}$

96-edge boundary / 64-vertex bulk; $P^{1.5} \leftrightarrow 96/64$ (C6, convergent, 35%). Unchanged at V6.

Separation Bound · preconditions explicit

$$I(S; M \mid FP) < \varepsilon^* \qquad \text{(load-bearing wall)}$$

Theorem (95%): Conditional independence $\implies$ additive MI bound $\implies R_{\max} < 1$.

Precondition 1 (non-collusion): $I(Y_S; Y_M \mid X) = 0$ and no third channel carries the inter-agent residue. **Precondition 2:** capacities against a stated adversary class. Grounding: Wyner (1975) equivocation; Fano converse; Leung-Yan-Cheong and Hellman (1978); Bayes-capacity; Geiger and Kubin.

Amnesia: $\varepsilon_{\text{amnesia}} < \varepsilon_{\text{policy}}$ (C17, quantitative at V6 via the compounding bound below). Topology > policy.

Betweenness centrality: $C_B(v) = \sum_{s,t} \sigma(s,t|v)/\sigma(s,t)$ (Brandes, 2001). The $\perp$ is the node with maximal betweenness in the trust graph.

The Reconstruction Ceiling · Proven, conditional regime

$$R_{max} = \frac{C_S + C_M}{H(X)} < 1 \qquad P_e \geq 1 - R_{max}$$

Both hold inside Preconditions 1 and 2; time-indexed per R(t). Outside the regime, the compounding bound governs:

$$I(O_N; S_1, \ldots, S_N) \;\leq\; \sum_i 2^{N-i} \varepsilon_i \;=\; (2^N - 1)\varepsilon \;\; \text{(uniform)}$$

(Asif-Amiri Theorem 4.1.) Amnesia separation breaks the chain and caps at $N\varepsilon$ (C83, 55%). The policy-to-amnesia gap is exponential-to-linear in N: at N=2, 3ε versus 2ε; at N=5, 31ε versus 5ε. C17 is hereby quantitative.

Algebraic Foundation · with the bridge

$\mathcal{L} = (\mathbb{Z}/64\mathbb{Z},\ +,\ \times) \qquad D_{64} = \langle \text{neg}, \text{bnot} \mid \text{neg}^2 = \text{bnot}^2 = 1,\ (\text{neg} \circ \text{bnot})^{64} = 1$

Op	Formula	Agent	Function
neg	$(64 - x) \bmod 64$	Swordsman	Boundary. Additive inverse.

Op	Formula	Agent	Function
bnot	$63 - x$	Mage	Projection. Bitwise complement.
$\text{neg} \circ \text{bnot}$	$x + 1 = \text{succ}(x)$	First Person	The step forward. ■

PRISM coordinates: $\text{blade}(x) = (\delta, \sigma, s)$: datum, stratum (Hamming weight), spectrum.

Pascal: $\{1, 6, 15, 20, 15, 6, 1\}$. Tiers: Null(0) / Light(1–2) / Heavy(3–4) / Dragon(5–6).

Six dimensions: Protection, Delegation, Memory, Connection, Computation, Value.

Hexagram: $[d_1 \ldots d_6] \to$ 64 I Ching states. Blade 63 = 111111 = Qian (The Creative).

All of the above carries unchanged. V6 adds: ARCH-1 enters the lineage, $\Sigma := \mu S.\,(\beta \vee \Omega(S, S))$ with activation ρ (C26-C29); operational layer ARCH-1R/T (C72-C76): ternary $\tau : T \to \{+, 0, -\}$, latent ≠ obstructed, $T = \text{orbit}(\rho, G)$. Bridge (C85, 40%): the axes and the triadic coordinates are one primitive; **the gap is β** (the recursion branches share only the First Person). Predictions: bnot-pairs invert all three axes; stratum-3 carries no dominant axis.

Operational Cycle

$$\text{cycle}(x) = \text{succ}(x) = \text{neg}(\text{bnot}(x))$$

Observe → Boundary → Project → Return:

Stage	Operation	Agent	Ceremony
Observe	$\text{id}(x)$	First Person	Sun · disclosure
Boundary	$\text{neg}(x)$	Swordsman	Gap · silence
Project	$\text{bnot}(\text{neg}(x))$	Mage	Moon · reflection
Return	$\text{succ}(x)$	Composition	Recursion

$T_\int(\pi) = 1 + \beta \sum_i \mathrm{cycle}(\mathrm{step}_i)$. Progressive trust: Understanding $\to$ Constellation $\to$ Blade $\to$ Runecraft.

Amnesia Protocol · the obstruction upgrade

Definition (reachability statement, V5.4): Structural amnesia w.r.t. origin O if no permitted operation sequence can reconstruct O.

ZK: completeness (output demonstrates), soundness (unique configuration), zero-knowledge (origin hidden).

Implementation: process boundary. Cosmological: Moon's orbit. Runecraft: Ed25519 key burned on close.

Obstruction statement (C86, 30%): Grade-2 forgetting $\iff$ the obstruction class to gluing local agent views into a global witness of O is non-vanishing; Grade-1 $\iff$ vanishing class, withheld gluing data. Terminal obstruction = structural loss of β (C73, 50%). Consequence if C86 holds: **amnesia is the only term whose security is independent of t.** Falsification: any view-composition recovering a Grade-2-forgotten O.

Forge Cryptography · with the accumulator

Property	Method
Content addressing	SHA-256
Tamper evidence	Hash chain
Pre-evocation lock	Commitment scheme
Identity binding	Ed25519 (Mage, held)
Bilateral binding	Dual Ed25519 (Mage held + Swordsman burned)

Moon phase: stratum $\longrightarrow$ visibility ratio. 0 = New Moon, 6 = Full Moon.

V6 adds the City Key trust recursion as IVC (C87, 50%): Key = folded accumulator · trust tasks = step circuits · Charge = folding step · V63 = attested invariant. Substrate line: Nova, HyperNova, MicroNova, LatticeFold (post-quantum hedge). Presence regime 1: 🔗 is non-transferable, non-attesting local color; upgrade ladder: witness co-signing, elapsed-time proofs.

The Temporal Thread

$$X_b + Y_b > Z_b, \qquad Z_b = t^*$$

Behavioural Mosca (C49, 70%) closed onto the moving ceiling. Existence-Leak (C81, 70%): $I(\text{feasibility}; \text{method}) > 0$; floor: leakage-resilient ZK with $\lambda < 1$ impossible (Garg-Jain-Sahai). Discount (C84, 50%): $Z_b' = Z_b - D(a)$ on every public feasibility attestation. Countermeasure chain (C18-C21, 10-30%): $|\pi(t) - \pi'(t)| \gtrsim |\delta_0| e^{\lambda t}$, λ unmeasured. Taxonomy: ages progressively (C47, 50%). Trust edges: half-life from inscription (C30-C33).

The Geometry of the Gap

Stella octangula: no golden ratio (volumes rational: tetrahedron 1/3, octahedral core 1/6, compound 5/12 of the cube); C1 beside the figure is resonance, not derivation. φ lives in the lattice: $\delta(38) = 38/63 = 0.60317$ vs $1/\varphi = 0.61803$, gap 2.4% (C54). Parity cube (C88, 30%): the two tetrahedra are the even/odd parity classes of $\{0,1\}^3$; $\{0,1\}^6 = \{0,1\}^3 \times \{0,1\}^3$. Octahedral gap (C89, 30%): the intersection is the locus of the conditional-independence bound; β, the conditioning, and the core are three readings of one thing.

Conjectures

Numbering authority: `CONJECTURE_REGISTER_V6.md` , head C89. Bands: C1-C17 core · C18-C37 V6 series · C38-C46 bound collection · C47-C55 coherence (C51-C55 occupied) · C56-C66 City · C67-C71 Horizon (pinned v1.8.0) · C72-C80 renumbered at the register lock · C81-C89 registered during the V6 runs. Aliases: C46↔C32 · C60↔C48 · C61↔C49 · CM-C47↔C85.

Core band, V5.4 baseline (confidences as of April 2026; the register supersedes where they differ):

ID	Claim	Confidence
C4	96/64 discrepancy	**RESOLVED**
C6	$P^{1.5} \leftrightarrow 96/64$ structural	**CONVERGENT** 35%
C7	Three-axis multiplicative	30%
C11	ρ amplifies + indicates maturity	55%
C12	Hexagram encoding	60%
C13	Bilateral witness quantum-resistant	65%
C14	$\Phi_a \cong D_{2n}$	75%
C15	$T_\int \cong$ resolution pipeline	65%
C16	Betti number trust invariants	25%
C17	Amnesia > policy separation	60% (quantitative at V6 via C83)
C18	Strange attractor dynamics ($\lambda > 0$)	25%
C19	ρ = Lyapunov divergence	20%
C20	Three axes couple as Lorenz variables	30%
C21	Fractal sovereignty dimension	10%

Proven Results · scoped

V5.4 baseline (95%):

1. Additive MI bounds from conditional independence
2. Reconstruction ceiling $R < 1$ under budget constraints
3. Error floor via Fano's inequality
4. Graceful degradation under partial compromise

5. Ring algebra $\mathbb{Z}/(2^6)\mathbb{Z}$ substrate
6. Two-extension autonomy axiom (separate processes)
7. Pretext DOM-free measurement as privacy primitive

V6 scoping: Additive leakage $I(X;Y_S,Y_M) = I(X;Y_S) + I(X;Y_M)$ at 95% **inside Precondition 1 only**. Separation bound and ceiling: Proven, conditional regime. Worked instances of the drift (2026): Orchard (flaw 2022-05, found 2026-05-29, fixed block 3,364,600 on 2026-06-03); Schrottenloher (eprint 2026/1128, 2026-06-02, ~2 months after the ZK attestation).

Version Lineage

Version	Date	Core Addition
V1	2024	$P \cdot C \cdot Q \cdot S$
V2	Oct 2025	$+e^{-\lambda t}$, network effects
V3	Nov 2025	$+R(d), M, \Phi$
V4	Feb 2026	$+\Sigma, A(\tau), T(\pi)$
V5	Feb 2026	$+$ three-axis Φ, holographic bound, $T_\int$
V5.1	Mar 29	$+\rho$, bilateral witness (C11–C13)
V5.2	Mar 31	$+D_{2n}$, PRISM (C14–C16)
V5.3	Apr 4	$+$ operational cycle, amnesia (C17)
V5.4	Apr 10	Consolidated. C18–C21. Full references.
V5.5	2026	Sublayer: attachment architecture
V6	**Jun 2026**	**The gathering turn**

V6 in full: R(t), t*; preconditions and external provenance; the exponential-to-linear gap; the Existence-Leak law; the bridge; obstruction amnesia; the Key as accumulator; the honest geometry; the unified register; unified V6 labels across all canon papers.

References

V6 grounding: Wyner 1975 · Fano 1961 / Cover-Thomas · Leung-Yan-Cheong & Hellman 1978 · Csiszár & Körner 1978 · Asif & Amiri arXiv:2603.05520 · Patil et al. arXiv:2509.14284 · AgentLeak arXiv:2602.11510 · Garg-Jain-Sahai · Schrottenloher eprint 2026/1128 · Blanco-Romero et al. arXiv:2603.01091 · Bakhta 2026 · Nova / HyperNova / MicroNova / LatticeFold · Mosca 2018 · Bergstra & Burgess 2019 · full list: privacy_value_v6.md §32.

V5.4 baseline references, carried: Shannon (1948). Fano (1961). Cover & Thomas (2006). Bergstra & Burgess (2019). Susskind (1995). McGilchrist (2009). Groth (2016). PLONK (2019). Nova (2022). Dwork & Roth (2014). Brandes (2001, 2008). Branco et al. (2025). Babbush et al. (2026). Cain et al. (2026). IEEE 7012-2025. UOR Foundation (2026). Hope & Ludlow (2023). Weyl & Tang (2023).

Narrative corpus (V5.4 record): The First Person Spellbook (31 acts, v10.0.0, CLOSED). Blog: sync.soulbis.com (Parts 0–5). Six grimoires: First Person, Zero Knowledge, Canon, Parallel Society, Plurality, City of Mages (Second Person). The Second Person Spellbook opened 2026-05-08 as the bound collection at `tomes/` ; Tome IV (Witnessing · 5 acts) closed; Tome V (Crafting) open at the City of Mages on Drake Island.

Full spec: `privacy_value_v6.md` . Companion: `pvm_v6_companion_guide.md` .

Docs: github.com/mitchuski/agentprivacy-docs. Forge: spellweb.ai. Training: agentprivacy.ai. Trust: bgin.ai.

The boundary is always enough. Peer review invited: mage@agentprivacy.ai

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